

Tutorial-III

Q1 Find the inverse of the following matrices by adjoint method:

(a) $\begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$, (b) $\begin{bmatrix} 2 & 3 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$, (d) $\begin{bmatrix} 1 & 0 & -1 \\ -2 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$

(e) $\begin{bmatrix} 24 & -4 & -4 \\ -4 & 24 & -4 \\ -4 & -4 & 4 \end{bmatrix}$

Q2 Using matrix inversion method solve the following equations.

(a) $x+y+z=3$, $x+2y+3z=4$, $x+4y+9z=6$

(b) $x+3y+3z=12$, $x+4y+4z=15$, $x+3y+4z=13$

(c) $4x-3y+z=1$, $x+4y-2z=10$, $2x-2y+3z=4$

(d) $y+z=5$, $3x+z=6$, $y-x=1$

(e) $x+y+z=3$, $3x-2y+3z=4$, $5x+5y+z=11$